Example 7.11 The dimensions of a fish tank are 3.8 m × 2.5 m × 1.6 m. How many litres of water it can hold?



Fig. 7.23

Solution

Length of the fish tank l =3.8 m

Breadth of the fish tank b =2.5 m , Height of the fish tank h =1.6 m

Volume of the fish tank $= l \times b \times h$

$= 3.8 \times 2.5 \times 1.6$

$= 15.2\ m^3$

$= 15.2 \times 1000\ litres$

$= 15200\ litres$

Note

A few important conversions

1 cm^3 =1 ml, 1000 cm^3 =1 $litre$,

1m^3 =1000 $litres$

Example 7.12 The dimensions of a sweet box are 22 cm × 18 cm × 10 cm. How many such boxes can be packed in a carton of dimensions 1 m × 88 cm × 63 cm?

Solution

Here, the dimensions of a sweet box are Length (l) = 22cm, breadth (b) = 18cm, height (h) = 10 cm.

Volume of a sweet box = $l \times b \times h$

$= 22 \times 18 \times 10\ cm^3$

The dimensions of a carton are

Length (l) = 1m= 100 cm, breadth (b) = 88 c

height (h) = 63 cm.

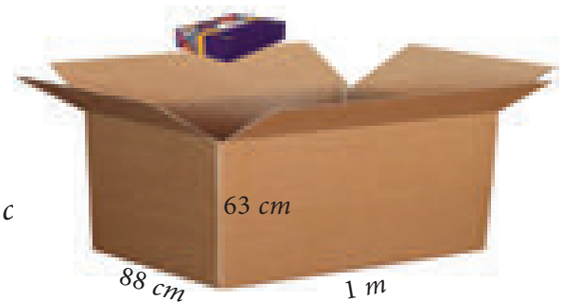


Fig. 7.24

Volume of the carton = $l \times b \times h$

$= 100 \times 88 \times 63\ cm^3$

The number of sweet boxes packed $= \dfrac{\text{volume of the carton}}{\text{volume of a sweet box}}$

$= \dfrac{100 \times 88 \times 63}{22 \times 18 \times 10}$

$= 140$ boxes

7.5.2 Volume of a Cube

It is easy to get the volume of a cube whose side is a units. Simply put $l = b = h = a$ in the formula for the volume of a cuboid. We get volume of cube to be a^3 cubic units.

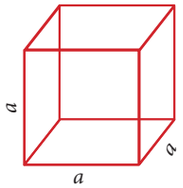


Fig. 7.25

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